Treatment in the Natural Cycle

Natural cycle treatment assumes: Normal ovulatory function with spontaneous (unstimulated) ovulation; At least one patent fallopian tube; Normal uterine cavity. Treatment options in the natural cycle encompass Timed coitus, Cervical insemination, Intrauterine insemination (IUI) and assisted reproductive technologies (ART). Cervical insemination in the natural cycle may be beneficial in cases involving sexual dysfunction. Intrauterine insemination may be useful in cases involving cervical trauma (e.g., cervical ablation, following a wide cervical cone biopsy). There is no evidence that absent sexual dysfunction or cervical trauma, natural cycle (i.e., no ovarian stimulation) IUI has any benefit over appropriately timed heterosexual intercourse. (Helmerhorst, 2005; ASRM, 2020). Natural cycle IUI may be considered in the setting of donor insemination when no other infertility factor is present.

Male Factor Infertility

World Health Organization Reference Limits for Human Semen Characteristics: Semen Parameter: One-Sided Lower Reference Limit (Fifth Centiles With 95% Confidence Intervals): Semen Volume 1.4 mL (1.3-1.5) Total Sperm Number 39 million per ejaculate (35-40) Sperm Concentration 16 million/mL (15-18 million/mL) Vitality 54% Live (50-56%) Progressive Motility 30% (29-31%) Total Motility (Progressive + Non-Progressive) 39% (40-43%) Morphologically Normal Forms 4.0% (3.9-4.0)

- Mild Male Factor: abnormalities in the semen analysis where the sperm concentration is ≥10 million/mL but <15 million/mL and/or progressive motility is ≥ 30% but <40% or ≥ 5 million total motile sperm
- Moderate Male Factor: abnormalities in the semen analysis where the sperm concentration is ≥5 million/mL but <10 million/mL and/or progressive motility is ≥ 25% but <30%
- Severe Male Factor: abnormalities in the semen analysis where the sperm concentration is <5 million/mL or sperm preparation techniques result in a sperm concentration of <1 million motile sperm/mL (Schlegel, 2020)
- Isolated teratospermia is considered a male factor when there is <2% normal morphology on at least two semen analyses 1-4 weeks apart.

Surgical Sperm Aspiration

Surgical sperm aspiration is the surgical removal of sperm to obtain high quality sperm in adequate numbers to be used in assisted reproductive technology cycles and/or cryopreservation. Approximately 5%-10% of males evaluated for infertility are azoospermic. (Schlegel,1997; Schlegel, 1999)

Elective Single Embryo Transfer (eSET)

Assisted reproductive technology (ART) poses a major risk of multiple pregnancy and birth that is associated with adverse maternal and infant outcomes. The principal reason behind the large number of multiple pregnancies after invitro fertilization (IVF) is the practice of transferring more than one embryo within the uterus to maximize pregnancy rates. (ASRM, 2012; Criniti, 2005; Pandian, 2009). Twin pregnancies and higher order gestations are associated with an increased risk of Preeclampsia,